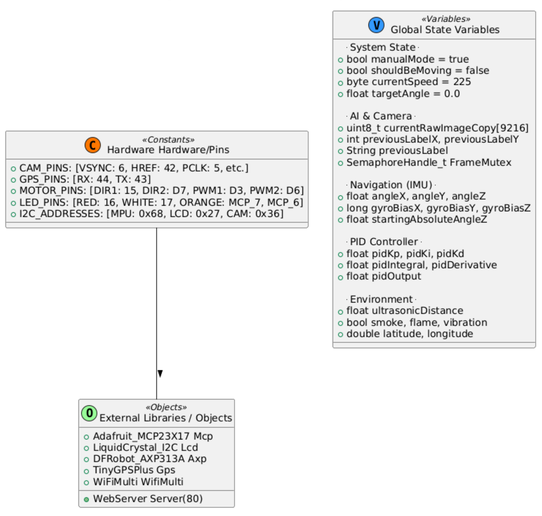
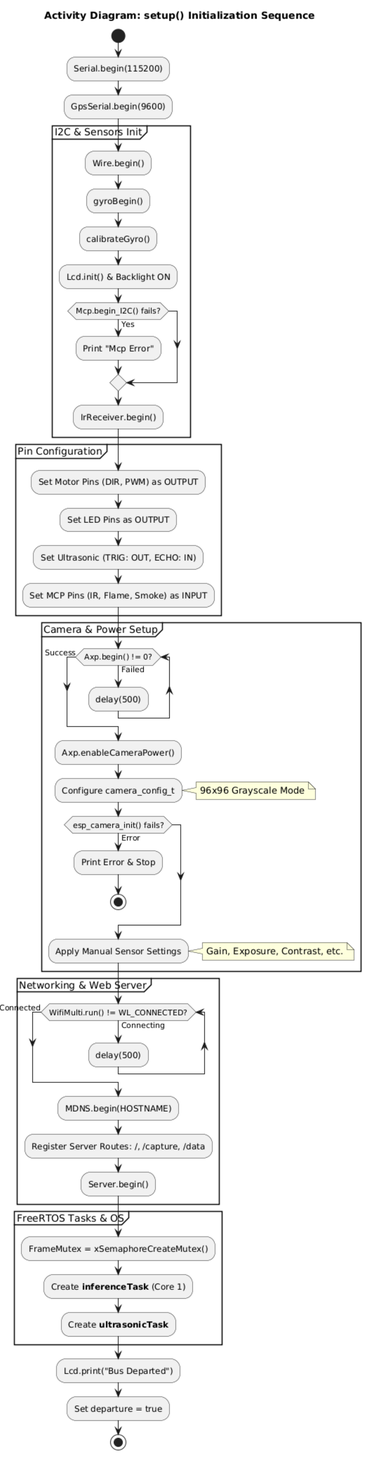
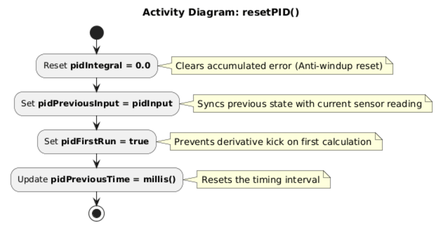
Robot Car System Architecture (Global Scope & Constants)



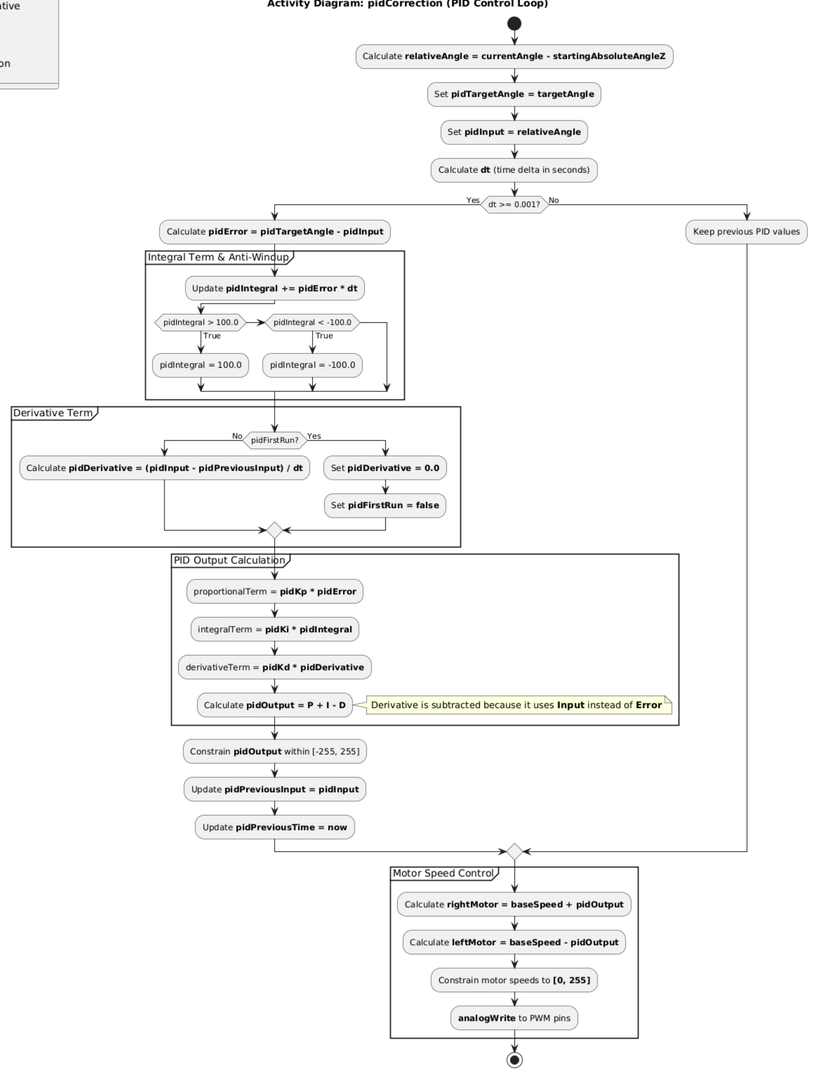
Activity Diagram: setup() Initialization Sequence



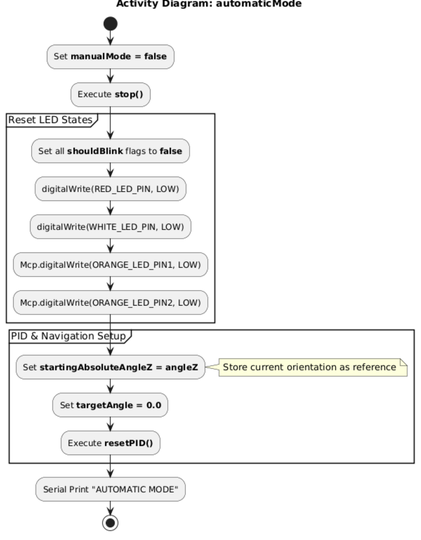
Activity Diagram: resetPID()



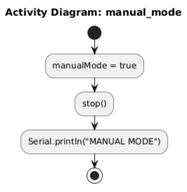
Activity Diagram: pidCorrection (PID Control Loop)



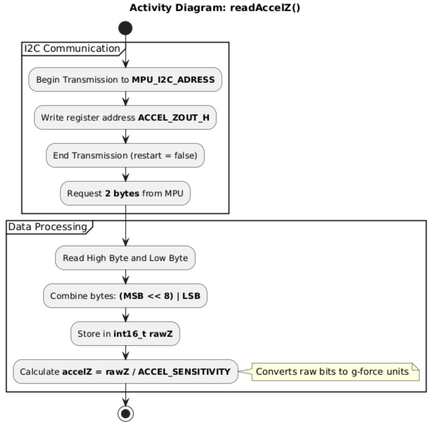
Activity Diagram: automaticMode



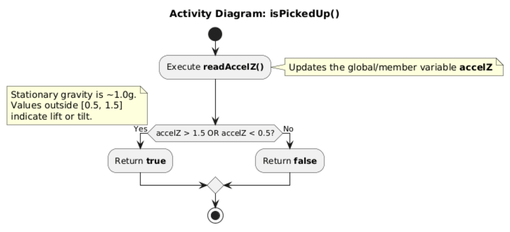
Activity Diagram: manual_mode



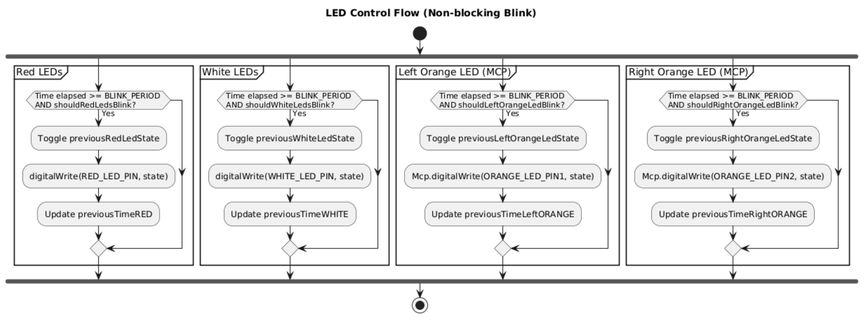
Activity Diagram: readAccelZ()



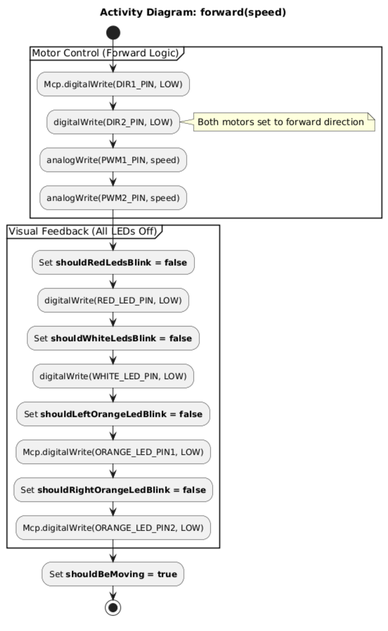
Activity Diagram: isPickedUp()



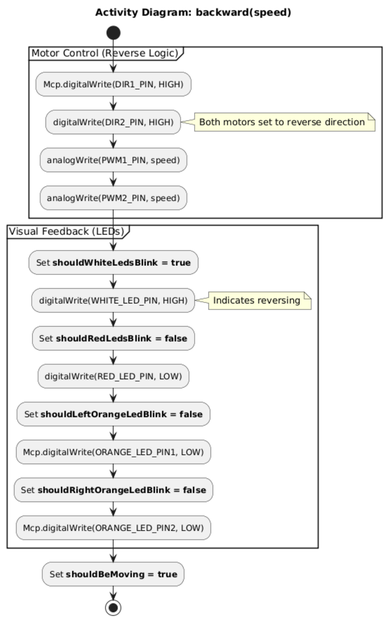
LED Control Flow (Non-blocking Blink)



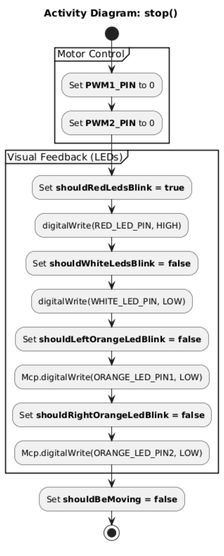
Activity Diagram: forward(speed)



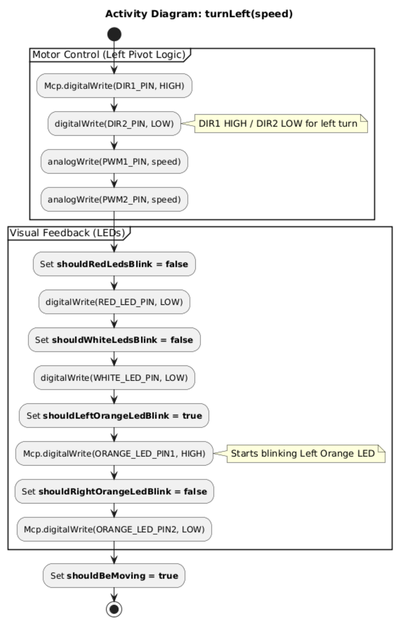
Activity Diagram: backward(speed)



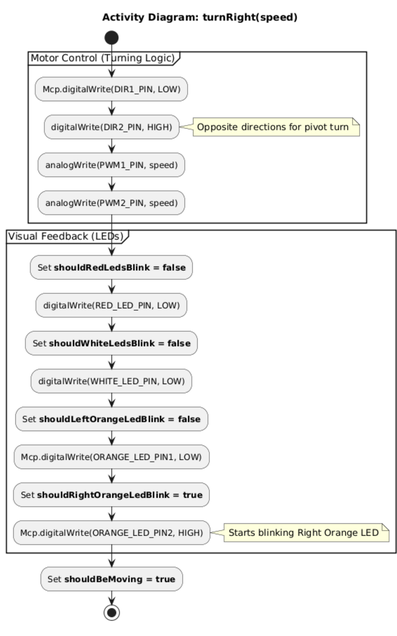
Activity Diagram: stop()



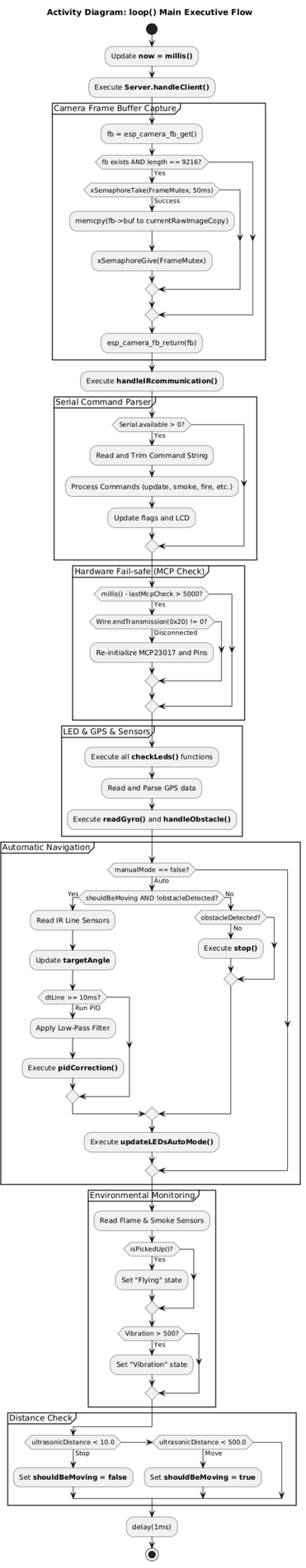
Activity Diagram: turnLeft(speed)



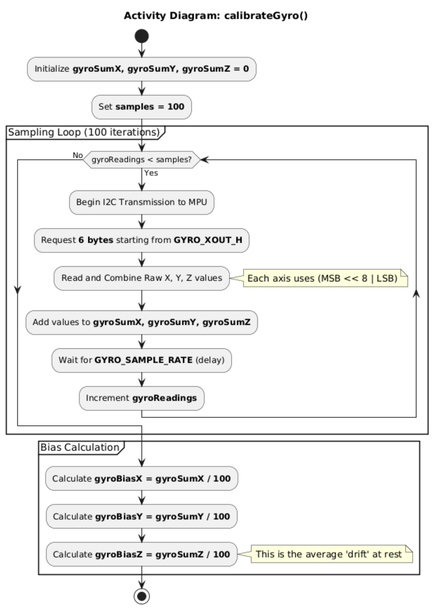
Activity Diagram: turnRight(speed)



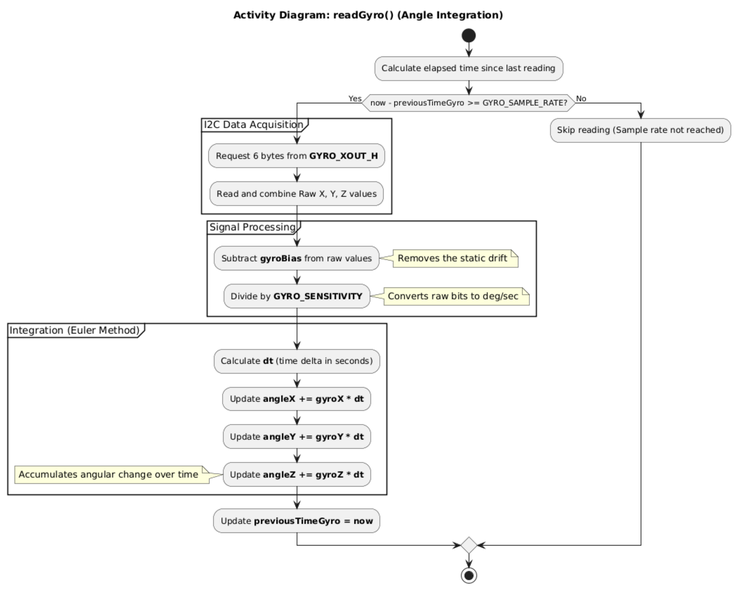
Activity Diagram: loop() Main Executive Flow



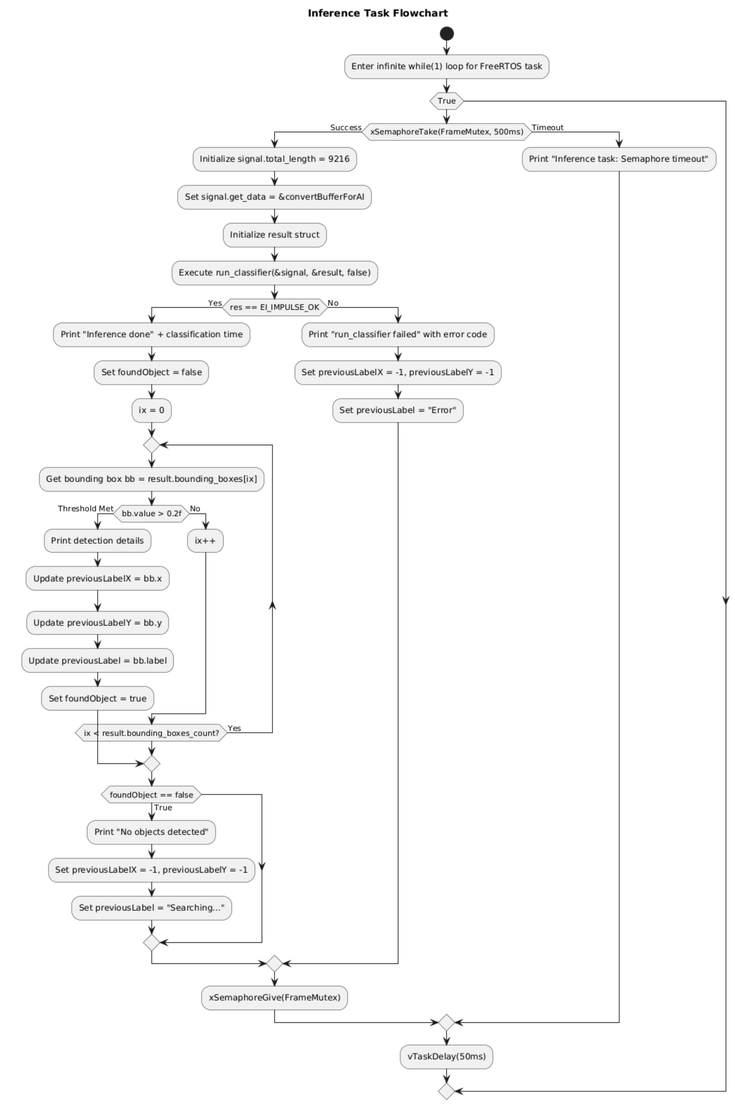
Activity Diagram: calibrateGyro()



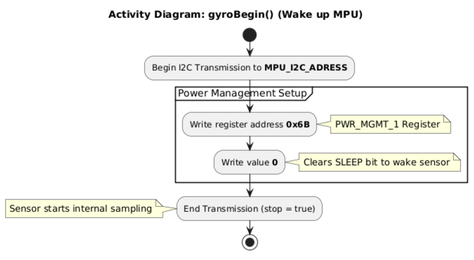
Activity Diagram: readGyro() (Angle Integration)



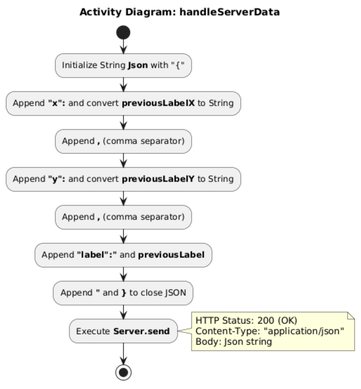
Inference Task Flowchart



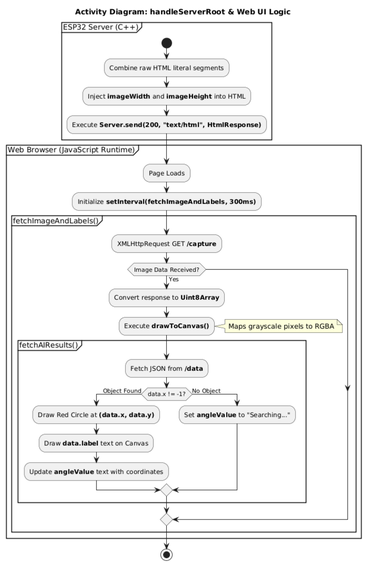
Activity Diagram: gyroBegin() (Wake up MPU)



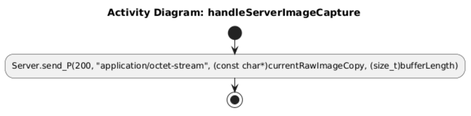
Activity Diagram: handleServerData



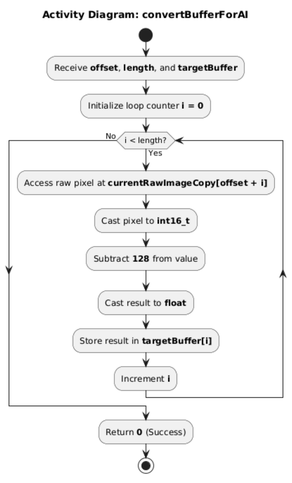
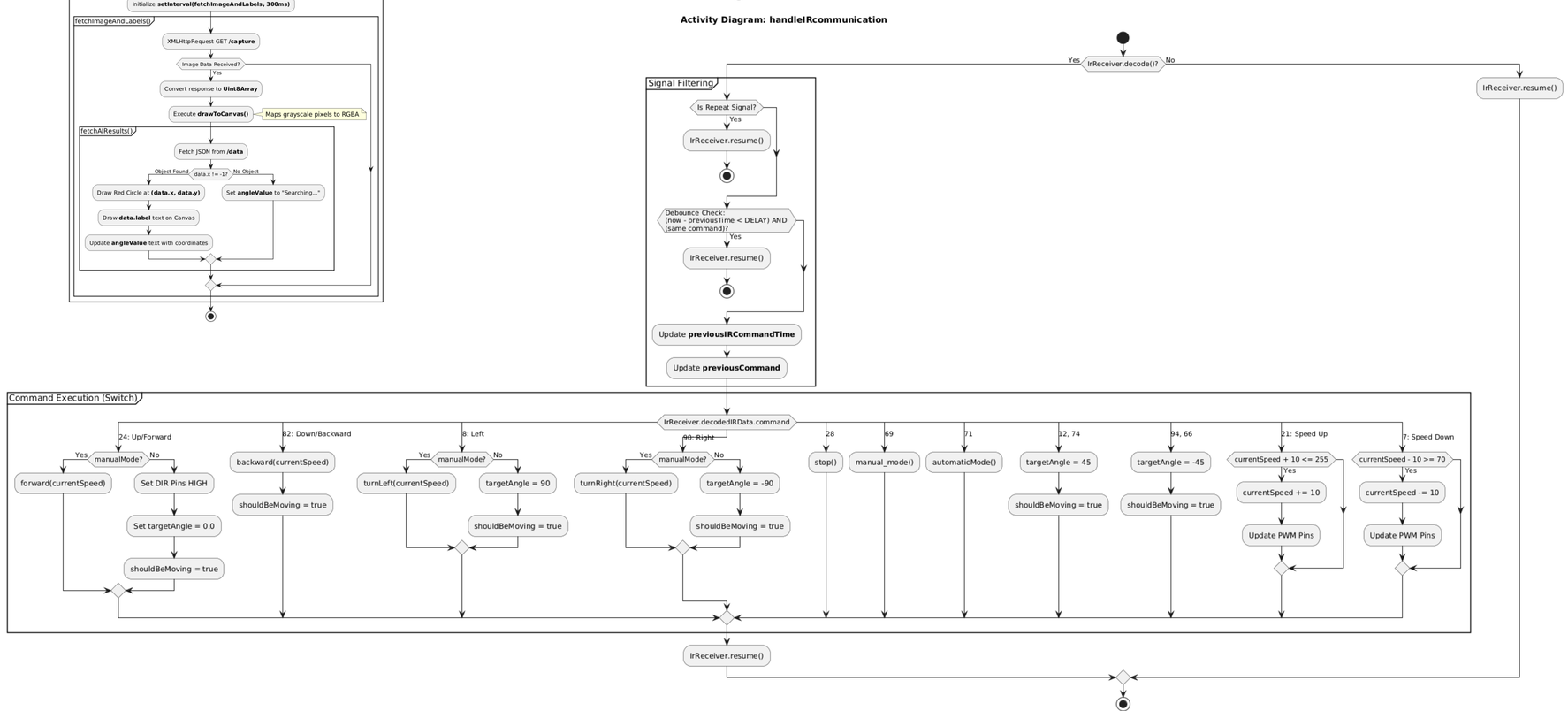
Activity Diagram: handleServerRoot & Web UI Logic



Activity Diagram: handleServerImageCapture



Activity Diagram: convertBufferForAI



Activity Diagram: handleIRcommunication